from: Stromberg, K., The Banach-Tarski paradox,
Amer. Math. Monthly 86 (1979), 151–161

Prop Let ρ be a rotation. Then we have the following.
(i) The image ρ of any line is a line:
$\rho(b+tc) = \rho(b) + t\rho(c)$ for all $b, c \in \mathbb{R}^3$ and $t \in \mathbb{R}$.
(ii) Inner products are preserved by ρ;
if $x, x' \in \mathbb{R}^3, \rho(x) = y$ and $\rho(x') = y'$, then

$$\sum_{i=1}^{3} y_i y_i' = \sum_{j=1}^{3} x_j x_j'.$$

(iii) Distances are preserved by ρ: if $x \in \mathbb{R}^3$, then $|\rho(x)| = |x|$.
(iv) If $\rho \neq \iota$, then the set $A = \{x \in \mathbb{R}^3 : \rho(x) = x\}$ is a line through the origin: there is a p in $\mathbb{R}^3$ such that $A = \{tp : t \in \mathbb{R}\}$ and $|p| = 1$. We call A the axis of ρ.
(v) If q is any point of $\mathbb{R}^3$ having the two properties of p in (iv), then $q = p$ or $q = -p$. We call p and $-p$ the poles of ρ.

Thm A If $cos\theta$ is a transcendental number, then each element of G other than ι has exactly one expression as a reduced word in the letters $\Phi, \psi,\ \psi^2$. That is, if
$(i)\ \ \sigma_1\sigma_2\cdots\sigma_n = \rho_1\rho_2 ... \rho_m$
where each side of this equation is a reduced word, then $m = n$ and $\sigma_j = \rho_j$ for $1 \leq j \leq n$.

Def Φ rotates $\mathbb{R}^3$ by 180° about the line in xz-plane whose equation is $x\cos\frac{1}{2}\theta = z\sin\frac{1}{2}\theta$.

$$\Phi = \begin{pmatrix} -\cos\theta & 0 & \sin\theta \\ 0 & -1 & 0 \\ \sin\theta & 0 & \cos\theta \end{pmatrix}$$

choose and fix θ such that $\cos\theta$ is transcendental; e.g., $\theta = 1$.

Def ψ rotates $\mathbb{R}^3$ by 120° about the z-axis.

$$\psi = \begin{pmatrix} -1/2 & -\sqrt{3}/2 & 0 \\ \sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Def If an element $\rho \in G$ is expressed in its unique way as a reduced word as in (2), we call n the length of ρ and we say that ρ_1 is the first letter of ρ or that ρ begins with ρ_1. We write $l(\rho) = n$ and $l(\iota) = 0$.

Thm B There exists a partition $\{G_1, G_2, G_3\}$ of G into three nonvoid subsets such that for each ρ in G we have
(i) $\rho \in G_1 \iff \Phi\rho \in G_2 \cup G_3 3$,
(ii) $p \in G_1 \iff \psi\rho \in G_2$,
(iii) $p \in G_1 \iff \psi^2\rho \in G_3$.

Lem If P is any countable subset of S, then there exists a countable set Q and a rotation ω such that $P \subset Q \subset S$ and $\omega(Q) = Q \setminus P$.

Thm C (AoC) There exists a partition $\{P, S_1, S_2, S_3\}$ of the unit sphere $S = \{x \in \mathbb{R}^3 : |x|^2 = x_1^2 + x_2^2 + x_3^2 = 1\}$ into four subsets such that
(i) P is countable, (ii) $\theta(S_1) = S_2 \cup S_3$,
(iii) $\psi(S_1) = S_2$, (iv) $\psi^2(S_1) = S_3$.

Thm D There exists a partition $\{T_j : 1 \leq j \leq 10\}$ of the unit sphere S into ten (disjoint) subsets and a corresponding set $\{\rho_j : 1 \leq j \leq 10\}$ of rotations such that $\{\rho_j(T_j) : 1 \leq j \leq 6\}$ is a partition of S into six subsets and $\{\rho_j(T_j) : 7 \leq j \leq 10\}$ is a partition of S into four subsets. Moreover, we can take T_7, T_8, and T_9 to all be rotates of S_1 and take T_1, T_2, T_3, and T_{10} to all be countable.

Def We shall say that two subsets X and Y of $\mathbb{R}^3$ are piecewise congruent and we write $X \sim Y$ if, for some natural number n, there exist a partition $\{X_j : 1 \leq j \leq n\}$ of X into n subsets and a corresponding set $\{f_j : 1 \leq j \leq n\}$ of rigid motions such that $\{f_j(X_j) : 1 \leq j \leq n\}$ is a partition of Y. In case X is piecewise congruent to a subset of Y, we shall write $X \lesssim Y$.

Thm E There exists a partition $\{B_k : 1 \leq k \leq 40\}$ of the closed unit ball B into forty subsets and a corresponding set $\{rk : 1 < k \leq 40\}$ of rigid motions such that $\{r_k(B_k) : 1 \leq k \leq 24\}$ partitions B into twenty-four subsets and $\{r_k(B_k) : 25 \leq k \leq 40\}$ partitions B into sixteen subsets.

Thm F For subsets X, Y, and Z of R^3 we have
(i) $X \sim X$,
(ii) $X \sim Y \implies Y \sim X$,
(iii) $X \sim Y$ and $Y \sim Z \implies X \sim Z$,
(iv) $X \sim X \implies X \lesssim Y$,
(v) $X \lesssim Y$ and $Y \lesssim Z \implies X \lesssim Z$,
(vi) $X \subset Y \implies X \lesssim Y$,
(vii) $X \lesssim Y$ and $Y \lesssim X \implies X \sim Y$.

Rem 1 The number 40 is not the smallest possible ...

Thm G If $A \subset R^3$ is a closed ball and if $A_1, A_2, \ldots, A_n$ are a finite number of translates of A, then

$$A \sim \bigcup_{j=1}^{n} A_j.$$

Rem 2 None of the three sets $S_{k'} = \{tx : x \in S_k,\ 0 < t \leq 1\}, 1 \leq k \leq 3$, can be Lebesgue measurable.

Rem 3 A poor analogue can be explicitly constructed (no Axiom of Choice) in the plane as follows. ...
The reason that this analogue is "poor" is twofold: X is both countable and unbounded.

Thm H (Banach-Tarski Paradox) If X and Y are bounded subsets of R^3 having nonvoid interiors, then $X \sim Y$.